IMO Shortlist 2013 — C1

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October 6, 2025

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Let n be a positive integer. Find the smallest integer k with the following property: Given any real numbers $a_1, \ldots, a_d$ such that $a_1 + a_2 + \cdots + a_d = n$ and $0 \le a_i \le 1$ for $i = 1, 2, \ldots, d$, it is possible to partition these numbers into k groups (some of which may be empty) such that the sum of the numbers in each group is at most 1.

Solution

The smallest such k is $2n - 1$. First we show that $k \ge 2n - 1$. Then we show that if the partitioning is possible for some $k > 2n - 1$ groups then the partitioning is possible for $k - 1$ groups also, establishing the result that $k = 2n - 1$ is indeed minimal.

Claim 1: $k \ge 2n - 1$.

Proof: Consider the case, $a_1 = a_2 = \cdots = a_d = n/d$. Let's say that each group can hold at most g numbers. We need the inequality

$$g \cdot \frac{n}{d} \le 1 \iff g \le \left\lfloor \frac{d}{n} \right\rfloor$$

and since we are minimising k we want g to be as large as possible and so we pick $g = \lfloor d/n \rfloor \ge 1$. We need to have $k \ge d/g$ groups, since we are splitting a collection of size d intro groups of size $\le g$, and so $k \ge \lceil d/g \rceil$. We can write $d = gn + r$ for integer $0 \le r < n$. Our inequality becomes,

$$k \ge \left\lceil \frac{gn + r}{g} \right\rceil = \left\lceil n + \frac{r}{g} \right\rceil$$

Since $g \ge 1$ and $r \le n - 1$ the expression on the left is at most $\lceil n + (n-1)/1 \rceil = 2n - 1$ which is achieved in the case $(g, r) = (1, n - 1)$, thus $k \ge 2n - 1$ as desired. □

Claim 2: If the partioning is possible for $k \ge 2n$ groups, it's also possible for $k - 1$ groups.

Proof: Represent the k groups by the sets $s_1, s_2, \ldots, s_k$ where, $S(i) \le S(j)$ for all $i < j$, without loss of generality, where $S(i) = \sum_{a \in s_i} a$. We claim that $S(1) + S(2) \le 1$. Assume the contrary, that $S(1) + S(2) > 1$. Then

$$n = (S(1) + S(2)) + \cdots + (S(2\lfloor k/2 \rfloor - 1) + S(2\lfloor k/2 \rfloor)) + \varepsilon_k > \lfloor k/2 \rfloor \ge n$$

where $\varepsilon_k \ge 0$ is $S(k)$ if k is odd and is 0 otherwise. But since $n > n$ is ridiculous we have found the desired contradiction. Now since $S(1) + S(2) \le 1$ we can combine them into 1 group and thus the partitioning is possible for $k - 1$ groups. □

By combining claim 1 and claim 2 we prove that the smallest such k is $2n - 1$. ■

Remark: This is my first ever IMOSL solve!